

llmXive Automated Review of CollectionLoRA: Collecting 50 Effects in 1 LoRA via Multi-Teacher On-Policy Distillation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The review focuses on the accuracy of factual claims and their alignment with the provided evidence (tables, figures, and text).

1. Overstated Performance Claims vs. Tabular Data In the Introduction (lines 45-48) and the Contributions section (lines 62-65), the authors claim that CollectionLoRA “surpasses independent single-task teachers in concept fidelity.” This claim is not fully supported by the quantitative data in Table 1 (`table/main_result.tex`).

- **DINO Score:** The single-task “Base” model achieves a DINO score of **0.611**, while the proposed “Ours” method achieves **0.600**. The proposed method is actually lower on this standard metric.
- **CLIP Score:** The scores are nearly identical (0.726 vs 0.727).
- **VSA Score:** The proposed method does achieve a higher VSA (4.380 vs 4.075).

While the authors introduce VSA to address limitations of DINO in stylized scenarios, the blanket statement that the method “surpasses” teachers in fidelity is misleading when a standard metric (DINO) shows a decline. The text should be nuanced to reflect that it surpasses teachers *on the proposed VSA metric* or *in terms of style alignment (CLIP)*, rather than making a general claim of superior fidelity that contradicts the DINO results.

2. Mathematical Inconsistency in Deployment Overhead Claim The Introduction (line 66) and the “Deployment Cost Analysis” section claim that at 180 effects, the method reduces deployment overhead to **0.5%** of the conventional paradigm.

- **Evidence:** Table 2 (`table/deploy_cost.tex`) provides data up to 150 LoRAs. At 150 LoRAs, the baseline storage is $2.2G \times 150 = 330G$. The proposed method storage is $2.2G \times 3 = 6.6G$.
- **Calculation:** $6.6/330 = 0.02$ or **2%**.
- **Discrepancy:** The claim of 0.5% is not derived from the storage numbers presented in the table. If the 0.5% figure includes routing latency or other factors, the text must explicitly define the “overhead” metric being used. As written, the claim appears mathematically inconsistent with the provided table data.

3. Exaggeration of “Training Failure” In Section 4.4 (lines 235-238), the authors state that applying vanilla DMD to the multi-teacher setting “causes the student distribution to collapse... resulting in training failure.”

- **Evidence:** Table 3 (`table/ablation.tex`), specifically Experiment (1), shows a configuration with PDSR but without the proposed TS and TA-FM components (effectively a baseline DMD approach with routing). This experiment yields a CLIP of 0.725 and a VSA of 2.756.

- **Analysis:** While the performance is lower than the full method, the model clearly trains and produces results (it does not “fail” in the sense of collapsing to noise or producing no output). The term “training failure” is an overstatement of the empirical evidence, which shows degradation rather than total collapse. The text should be revised to describe this as “significant performance degradation” or “suboptimal convergence” rather than “failure.”

4. Citation Context The citations for DMD (dmd1, dmd2) and Flow Matching (lipman2023flow-matchinggenerativemodeling) are accurate and support the methodological descriptions. However, the claim that “existing DMD-based methods are largely confined to single-teacher... settings” (Section 2.2, line 135) is a standard literature review claim and appears accurate based on the cited works. No issues found with the citation support for the method’s theoretical basis.

Recommendation: The authors should revise the claims in the Introduction and Section 4.4 to align strictly with the numerical evidence in Tables 1 and 3. Specifically, qualify the “surpassing” claim to specific metrics and correct the “0.5%” overhead figure or provide the calculation basis for it.

Action items.

- **[writing]** DINO Score: The single-task “Base” model achieves a DINO score of 0.611, while the proposed “Ours” method achieves 0.600. The proposed method is actually lower on this standard metric.
- **[writing]** CLIP Score: The scores are nearly identical (0.726 vs 0.727).
- **[writing]** VSA Score: The proposed method does achieve a higher VSA (4.380 vs 4.075). While the authors introduce VSA to address limitations of DINO in stylized scenarios, the blanket statement that the method “surpasses” teachers in fidelity is misleading when a standard metric (DINO) shows a decline. The text should be nuanced to reflect that it surpasses teachers *on the proposed VSA metric* or *in terms of style alignment (CLIP)*, rather than making a general claim of superior fidelity that contradicts
- **[writing]** Evidence: Table 2 (table/deploy_cost.tex) provides data up to 150 LoRAs. At 150 LoRAs, the baseline storage is $2.2G \times 150 = 330G$. The proposed method storage is $2.2G \times 3 = 6.6G$.
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- **[writing]** Discrepancy: The claim of 0.5% is not derived from the storage numbers presented in the table. If the 0.5% figure includes routing latency or other factors, the text must explicitly define the “overhead” metric being used. As written, the claim appears mathematically inconsistent with the provided table data.
- **3. Exaggeration of “Training Failure”** In Section 4.4 (lines 235-238), the authors state that applying vanilla DMD to the multi-teacher setting “causes the student distribution to collapse..
- **[writing]** Evidence: Table 3 (table/ablation.tex), specifically Experiment (1), shows a configuration with PDSR but without the proposed TS and TA-FM components (effectively a baseline DMD approach with routing). This experiment yields a CLIP of 0.725 and a VSA of 2.756.

- **[writing]** Analysis: While the performance is lower than the full method, the model clearly trains and produces results (it does not “fail” in the sense of collapsing to noise or producing no output). The term “training failure” is an overstatement of the empirical evidence, which shows degradation rather than total collapse. The text should be revised to describe this as “significant performance degradation” or “suboptimal convergence” rather than “failure.” 4. Citation Context The citations for DMD (dmd1

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a placeholder promotional graphic that fails to function as a scientific figure, containing no data, axes, or explanatory labels to support the paper’s claims.

Figure 2

Figure 2 is a qualitative comparison of DINO and VSA scores for ‘Bad’ and ‘Good’ cases, but it lacks a caption and axis labels, making the specific metrics and their significance unclear.

Figure 3

Figure 3 effectively illustrates the paper’s claim that VSA is a more robust metric than DINO for subject consistency. The visual examples clearly contrast a ‘Bad Case’ where DINO fails (high score, VSA=0) with ‘Good Cases’ where both metrics align, and the caption accurately describes this behavior.

Figure 4

Figure 4 is a clear and well-structured qualitative comparison that effectively supports the caption’s claims. The grid layout logically organizes different methods and failure modes, with row and column labels that are legible and sufficient to understand the visual evidence presented.

Figure 5

Figure 5 effectively demonstrates the zero-shot effect composition capability of CollectionLoRA. The layout clearly presents individual effects, input images, and the combined results, with legible labels that align perfectly with the caption’s description.

Figure 6

The figure displays a qualitative comparison grid of image editing results, but the caption is missing ('no caption'), and the filename suggests it should be an ablation study, creating a significant mismatch between the visual content and the provided metadata.

Figure 7

Figure 7 is missing its caption, and the visual content (a qualitative grid) appears inconsistent with the filename suggesting a quantitative ablation study on training steps.

Figure 8

The figure is visually clear with distinct lines and a readable legend, but the caption is too brief to stand alone, failing to explain the specific ablation components being compared.

Figure 9

Figure 9 is a schematic diagram comparing a 'Previous Method' to 'Ours' but is critically flawed by the absence of a descriptive caption. The current placeholder text '(no caption)' fails to explain the workflow, components, or the specific advantages of the proposed 'Collection LoRA' architecture shown in panel (b).

Figure 10

The figure provides a clear visual overview of the framework with a helpful legend for trainable/frozen components. However, the caption contains a typo in the loss function notation and a missing closing parenthesis.

Figure 11

The figure displays qualitative results for 'Standard DMD' and 'Reference Effect' but fails to visually label the specific ablation components (C2F-DO, trajectory anchoring, target simulation) described in the caption, creating a disconnect between the text and the visual evidence.

Figure 12

The figure attempts to illustrate the difference between Backward and Target Simulation but fails to visually represent the 'severe domain deviation' claimed in the caption, as the input images appear identical in both panels. Additionally, the specific roles of the 'real' and 'fake' images in the distillation process are not explicitly defined in the diagram.

Action items.

- **[fatal]** Figure 1: The image is a promotional graphic for a '50 in 1 Collection LoRA' product rather than a scientific figure; it lacks axes, data, or methodological context required for a preprint.

- **[science]** Figure 1: The caption describes the figure as a ‘Placeholder’ and defines ‘Customized image editing’ in a way that does not match the visual content, which is a collage of generated images without explanatory text or labels.
- **[writing]** Figure 2: The figure has no caption provided, making it impossible to understand the context of the ‘Bad Case’ and ‘Good Case’ labels or the specific meaning of the DINO and VSA scores shown.
- **[science]** Figure 2: The ‘Bad Case’ example shows a generated image with a score of DINO: 0.952 and VSA: 0, but without a caption or axis labels, it is unclear what specific metric or comparison is being visualized to support the claim of VSA’s superiority over DINO.
- **[fatal]** Figure 6: The rendered image is a qualitative grid of image editing results labeled ‘Source Image’, ‘Exp.(1)’ through ‘Exp.(5)’, and ‘Reference Effect’, but the caption explicitly states ‘(no caption)’ and the filename ‘zip_ablation_compare.pdf’ suggests it should be an ablation study; the content does not match the provided caption or the implied figure type.
- **[fatal]** Figure 7: The figure is rendered as a qualitative grid of images (showing cats and dogs in pink houses) but lacks a caption. The filename ‘zip_ablation_trainsteps.pdf’ implies a quantitative ablation study on training steps, yet the visual content does not match the expected format for such a study, and the missing caption prevents verification of the figure’s purpose.
- **[writing]** Figure 8: The caption ‘DreamSim Distance’ is insufficient; it fails to define the three ablation conditions (Baseline, Baseline + Target Simulation, Baseline + Target Simulation + TA-FM) shown in the legend, making the figure’s specific contribution unclear without external context.
- **[fatal]** Figure 9: The caption is explicitly marked ‘(no caption)’ and the filename ‘[zip_intro.pdf]’ suggests this is a placeholder or intro slide rather than a scientific figure. It lacks a descriptive caption explaining the diagram’s components (e.g., ‘Effect LoRA Bank’, ‘Query Routing’, ‘Collection LoRA’) or its purpose, making it impossible to interpret the figure’s scientific contribution.
- **[writing]** Figure 10: The caption contains a typo in the loss function name, writing ‘ L_{TA-FM} ’ instead of ‘ L_{TA-FM} ’ to match the label in panel (b).
- **[writing]** Figure 10: The caption text is missing a closing parenthesis after the description of the Effect Stream in section (b).
- **[science]** Figure 11: The caption claims the figure demonstrates the ‘Effectiveness of C2F-DO’ and compares ‘trajectory anchoring’ vs ‘target simulation’, but the image labels only show ‘Standard DMD’ and ‘Reference Effect’. The specific ablation components (C2F-DO, trajectory anchoring, target simulation) are not visually identified in the figure, making the caption’s technical claims unverifiable from the visual evidence.
- **[writing]** Figure 11: The caption refers to sub-figures (a) and (b), but the image lacks explicit (a) and (b) labels to distinguish the ‘Standard DMD’ results from the ‘Reference

Effect’ results.

- **[science]** Figure 12: The ‘Source Image’ (cat) and ‘Target Image’ (cat in shorts) are identical in both panels (a) and (b), yet the caption claims panel (a) suffers from ‘severe domain deviation’ while panel (b) does not. The visual evidence contradicts the claim that the inputs differ in domain deviation; the figure fails to demonstrate the specific heterogeneous setting described.
- **[writing]** Figure 12: The labels ‘real’ and ‘fake’ are applied to the intermediate images, but the diagram does not explicitly define which image corresponds to the ‘student’ or ‘teacher’ output, relying on the reader to infer the distillation flow from the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript exhibits a high density of domain-specific acronyms and jargon that are frequently introduced without explicit definition, creating barriers for non-specialist readers.

In the **Introduction (Section 1)**, the term **VLM** (Vision-Language Model) is used in the description of the Asymmetric Orthogonal Prompting strategy without being spelled out. Similarly, **DMD** (Distribution Matching Distillation) is introduced with its full name, but subsequent references to **DMD2** and the specific mechanics assume the reader already knows the distinction between the two versions without further elaboration in the intro.

In **Section 4 (Method)**, the paper introduces several new acronyms: **PDSR**, **AOP**, **C2F-DO**, **TA-FM**, and **TS**. While the full names are provided, the text immediately pivots to using the acronyms in equations and subsequent paragraphs (e.g., Eq. 6, 7, 8) without a clear “hereafter referred to as” convention. Specifically, **TS** (Target Simulation) and **TA-FM** are introduced in Section 4.4, but the text relies on these abbreviations heavily in the ablation study descriptions without re-stating the full terms. Additionally, **OOD** (Out-of-Distribution) appears in Section 4.2 without definition.

In **Section 5 (Experiment)**, the jargon load increases. **MLLM** (Multimodal Large Language Model) is used to describe the evaluation engine for VSA and BCR metrics but is never defined. **NFE** (Number of Function Evaluations) appears in Table 1 without explanation. The baseline **50-in-1 (FM)** uses **FM** (Flow Matching) as an acronym without defining it in the experimental setup, assuming the reader recalls it from the preliminaries or related work. **BCR** and **VSA** are introduced as metrics but the acronyms are not explicitly expanded in the text before being used in the results analysis.

Finally, in the **Supplementary Material**, terms like **AAC** (Automated Asymmetric Conditioning) appear in figure captions and text without definition in the main body or the appendix introduction. The reliance on **lightx2v** as a proper noun without context or definition also adds

to the opacity.

To improve accessibility, the authors should ensure every acronym is defined at its first occurrence in the main text (e.g., “Vision-Language Model (VLM)”) and consider using the full term or a brief parenthetical explanation when re-introducing these concepts in the experimental sections, rather than assuming continuity from the method section.

Action items.

- **[writing]** The manuscript exhibits a high density of domain-specific acronyms and jargon that are frequently introduced without explicit definition, creating barriers for non-specialist readers. In the Introduction (Section 1), the term VLM (Vision-Language Model) is used in the description of the Asymmetric Orthogonal Prompting strategy without being spelled out. Similarly, DMD (Distribution Matching Distillation) is introduced with its full name, but subsequent references to DMD2 and the specific mechani

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical consistency of the paper is generally strong in its problem formulation and the high-level motivation for the proposed framework. The transition from identifying the “Dilemma of Multi-Module Composition” (Section 4.1) to the “Proposed Paradigm” is well-reasoned. However, there are specific gaps in the logical derivation of certain claims and the precise definition of the optimization objectives that require clarification.

First, in Section 4.4 (Coarse-to-Fine Distillation Objective), the formulation of the total loss $\mathcal{L}_{\text{C2F-DO}}$ includes $\mathcal{L}_{\text{DMD_BS}}$ (Backward Simulation). The text describes this term as a “persistent regularizer” within the Effect Stream. However, Section 4.2 defines the General Stream as utilizing $\mathcal{L}_{\text{DMD_BS}}$ with the *frozen base model* as the teacher. The logical connection is ambiguous: does the $\mathcal{L}_{\text{DMD_BS}}$ term in the Effect Stream also use the base model (acting as a generalization anchor) or the specific effect teacher? If it uses the effect teacher, it effectively becomes a standard DMD loss, which the authors claim causes collapse in the early stages. If it uses the base model, the logic of why it is included in the *Effect* stream (which is supposed to learn the effect) needs explicit justification to avoid circular reasoning regarding the “distribution gap.”

Second, the claim of “zero-shot effect composition” in Section 5.2 relies heavily on the Asymmetric Orthogonal Prompting (AOP) strategy. The authors assert that AOP “encodes each effect into an orthogonal subspace of the prompt manifold,” enabling compositional activation. While this is a plausible hypothesis, the paper lacks the logical or empirical evidence to support the premise of orthogonality. There is no analysis of the prompt embeddings (e.g., cosine similarity between trigger words) or ablation showing that non-orthogonal triggers fail to compose. The conclusion that the emergent capability is *caused* by the orthogonality of the prompts is an

assumption not fully supported by the provided evidence, which only shows the *result* of the composition.

Finally, there is a minor logical tension regarding the deployment claims. The Introduction states the method “fundamentally resolves” routing latency. However, Section 5.1 admits that for 100-150 LoRAs, the method “reverts to VLM routing.” While the storage overhead is still reduced, the re-introduction of routing logic at scale suggests the solution is not a complete elimination of the routing problem but a mitigation up to a certain threshold. The text should be adjusted to reflect this nuance to maintain logical consistency between the abstract claims and the detailed experimental analysis.

Action items.

- **[science]** Clarify the teacher source for L_DMD_BS in the Effect Stream (Sec 4.4). Is it the base model (like General Stream) or the specific effect teacher? The current text implies it acts as a regularizer but lacks logical precision on the target distribution.
- **[science]** The claim that AOP creates ‘orthogonal subspaces’ enabling zero-shot composition (Sec 5.2) lacks supporting evidence. Provide empirical proof (e.g., embedding similarity) or remove the causal claim linking orthogonality to the emergent capability.
- **[writing]** The Introduction claims routing latency is ‘fundamentally resolved,’ yet Sec 5.1 states routing reverts for 100+ LoRAs. Reconcile this contradiction by clarifying the scaling threshold where routing is re-enabled.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding performance and scalability that slightly exceed the immediate support provided by the experimental data, particularly in the Introduction and Conclusion sections.

First, the claim that the method “surpasses independent single-task teachers in concept fidelity” (Introduction, Contribution 3) is not fully substantiated. Table 1 explicitly shows that the single-task “Base” model achieves a higher DINO score (0.611) compared to the proposed “Ours” method (0.600). While the authors introduce a new metric (VSA) where their method excels, asserting general superiority over teachers without qualifying the specific metrics where the teacher still outperforms the student is an over-claim. The text should be nuanced to state that the method achieves *comparable* or *superior* fidelity depending on the specific metric (e.g., VSA vs. DINO), rather than a blanket statement of surpassing teachers.

Second, the discovery of “zero-shot effect composition” (Introduction, Contribution 3; Section 4.3) is framed as a general capability. However, the evidence in Figure AB_Test only demonstrates the combination of two specific effects that were part of the training set. The paper does not

provide evidence that the model can compose arbitrary, unseen pairs of effects or generalize this compositional ability to effects not explicitly trained in the 50/180 set. The claim should be tempered to specify that this applies to the composition of *trained* effects, rather than implying a general zero-shot compositional reasoning capability for any effect.

Finally, the assertion that the framework scales to 180 effects “without catastrophic quality degradation” (Introduction, Contribution 3) glosses over the quantitative decline in performance. Table 3 shows a drop in CLIP score from 0.727 (50 effects) to 0.709 (180 effects), and the 180-effect model underperforms the single-task base model (0.724). While the model does not fail completely, describing a noticeable drop in alignment scores as “without catastrophic quality degradation” minimizes the trade-off. The authors should more accurately characterize this as “graceful degradation” or explicitly acknowledge the performance cost of extreme scaling, rather than implying the quality remains effectively unchanged.

Action items.

- **[writing]** Claiming the method ‘surpasses independent single-task teachers’ (Intro) is unsupported as Table 1 shows the Base model has a higher DINO score (0.611 vs 0.600). Qualify this to specify superiority only in VSA, not all fidelity metrics.
- **[writing]** The ‘zero-shot effect composition’ claim (Intro) implies generalization to unseen pairs, but Fig. AB_Test only shows composition of two specific trained effects. Restrict the claim to ‘composition of trained effects’ to avoid over-generalization.
- **[writing]** Stating the framework scales to 180 effects ‘without catastrophic quality degradation’ (Intro) minimizes the CLIP drop from 0.727 to 0.709. Rephrase to acknowledge the performance trade-off or use ‘graceful degradation’ instead.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a technical framework for distilling multiple visual effects into a single LoRA. From a safety and ethics perspective, the primary concerns relate to data privacy in the evaluation pipeline and the ethical oversight of human-subject research.

First, the evaluation of the Valid Subject Alignment (VSA) and Bad Case Rate (BCR) metrics relies heavily on the **Qwen-VL-Max-Latest** API (Appendix, “Prompt Design for Evaluation”). The manuscript describes sending 5,000 test image pairs to this external service. There is no mention of the data privacy implications of this process. Specifically, the authors must clarify whether these images are retained by the API provider for model training or if they are processed in a strictly ephemeral, privacy-preserving manner. Given the potential for sensitive content in image editing datasets, explicit disclosure of the data handling policy for external API calls is required to ensure compliance with data protection standards.

Second, the “User Study and Additional Qualitative Results” section in the Supplementary Material reports a study involving 10 professional evaluators. The text states that evaluators were “kept blind to the underlying methods” and asked to rate images on quality, consistency, and style. However, the manuscript lacks any statement regarding ethical oversight. Standard research ethics require that studies involving human participants obtain informed consent and, typically, approval from an Institutional Review Board (IRB) or an equivalent ethics committee. The absence of such a statement is a significant omission for a paper claiming human evaluation results.

Finally, the Asymmetric Orthogonal Prompting (AOP) strategy (Section 4.3) utilizes a Vision-Language Model to generate descriptive captions for the student model based on the training data. While the paper mentions using “internally constructed special-effect image pairs,” it does not explicitly address the provenance of these images regarding copyright or the presence of personally identifiable information (PII). If the training data includes images of real people or copyrighted artwork, the process of feeding them into an external VLM for prompt generation should be accompanied by a statement confirming that appropriate licenses were secured or that consent was obtained from subjects.

These issues are primarily matters of transparency and procedural compliance rather than fundamental flaws in the algorithmic safety of the model itself. Addressing them will ensure the research adheres to standard ethical guidelines for AI publication.

Action items.

- **[writing]** The evaluation protocol relies on the Qwen-VL-Max-Latest API (Appendix, ‘Prompt Design for Evaluation’) to calculate VSA and BCR metrics. The manuscript must explicitly disclose the data privacy policy regarding the 5,000 test images sent to this external API, confirming whether images are retained for model training or processed in a privacy-preserving manner.
- **[writing]** The ‘User Study’ section (Supplementary Material) states that 10 professional evaluators were hired. The manuscript must confirm that informed consent was obtained from these participants and that the study protocol was reviewed by an Institutional Review Board (IRB) or equivalent ethics committee, as is standard for human-subject research.
- **[writing]** The ‘Asymmetric Orthogonal Prompting’ (AOP) strategy uses a VLM to generate student prompts from training data (Sec 4.3). The authors should clarify if the training images used for this prompt generation contain any personally identifiable information (PII) or copyrighted material, and if so, how consent or licensing was handled.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the central claims of CollectionLoRA is generally robust in its experimental design but lacks necessary statistical depth to fully validate the scalability and metric reliability claims.

Sample Size and Replication: The evaluation protocol mentions 5,000 instructions per model (Sec 5.1), which is a strong sample size for image generation tasks. However, the paper does not report the number of independent training runs or seeds used to generate the results in Table 1 and Table 3. In distillation tasks, performance can be highly sensitive to initialization and random seeds. Without reporting mean and standard deviation across multiple seeds (e.g., $N \geq 3$), the observed differences (e.g., the 0.001 difference in CLIP scores between the full model and the baseline in Table 1) cannot be distinguished from random noise. The ablation study in Table 3 presents single-point estimates, making it impossible to assess the statistical significance of the component contributions.

Effect Sizes and Statistical Significance: The claim that the method scales to 180 effects with “competitive performance” (Sec 1, Table 4) relies on a CLIP score of 0.709 versus a baseline of 0.722. While the authors argue this is acceptable, the lack of confidence intervals or hypothesis testing (e.g., t-tests) leaves the magnitude of the degradation ambiguous. Is the drop from 0.727 to 0.709 statistically significant? The current presentation treats these as deterministic values rather than estimates with variance.

Metric Validity and Alternative Explanations: The introduction of the Valid Subject Alignment (VSA) metric is a key contribution, but its validity rests entirely on the Qwen-VL-Max-Latest MLLM (Appendix 7). The paper provides no evidence that this specific MLLM’s scoring aligns with human perception, nor does it report inter-rater reliability if multiple evaluators were used. If the MLLM has a bias towards certain styles or fails to detect specific types of “bad cases” (e.g., subtle semantic drift), the entire conclusion regarding “superior concept fidelity” could be an artifact of the evaluator’s limitations rather than the model’s actual performance. A human evaluation study (beyond the small user study in the appendix) validating the VSA scores against human ground truth is required to rule out this alternative explanation.

Training Dynamics: The training dynamics plots (Fig 10) show convergence but lack error bands. Given the complexity of multi-teacher distillation, the stability of the optimization is a critical claim. The absence of variance visualization in these plots makes it difficult to assess the robustness of the proposed Coarse-to-Fine Distillation Objective against different random seeds.

In summary, while the experimental setup is sound, the statistical reporting is insufficient to confirm that the observed improvements are robust and not due to chance or evaluator bias.

Action items.

- **[science]** The claim of scaling to 180 effects (Sec 1, Tab 4) lacks statistical rigor. The CLIP score drop from 0.727 (50 effects) to 0.709 (180 effects) is presented as ‘competitive’ without confidence intervals or significance testing against the baseline. Provide error bars or p-values to substantiate that the degradation is not statistically significant.
- **[science]** The VSA metric relies entirely on a single MLLM (Qwen-VL-Max-Latest) for evaluation (Sec 5.1, App 7). No inter-rater reliability or human validation of this automated metric is provided. Given the high stakes of the ‘Bad Case’ classification, report a human-

in-the-loop validation study on a subset of samples to confirm the MLLM’s alignment with human judgment.

- **[science]** The ablation study (Tab 3) shows Exp (3) achieving higher CLIP (0.736) than the full model (0.727), yet the full model is claimed as optimal. The text attributes this to ‘stability’ but does not provide a statistical test or a multi-run average to prove the full model’s performance is not a result of random variance or overfitting to the specific test set.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the manuscript requires strengthening to support the quantitative claims. While the experimental design is sound in its structure, the reporting of results lacks necessary statistical rigor.

First, the primary metric for subject consistency, Valid Subject Alignment (VSA), is derived entirely from a Multimodal Large Language Model (MLLM) API. The paper asserts that this metric is superior to DINO but provides no statistical evidence of the MLLM’s reliability. There is no report of inter-rater reliability (e.g., Cohen’s Kappa) between the MLLM and human annotators, nor are there confidence intervals for the VSA scores. Without quantifying the variance or error rate of the MLLM evaluator, the reported differences (e.g., 4.380 vs 4.150) may reflect evaluator noise rather than true model performance.

Second, the ablation study (Table 4) and scaling experiments (Table 5) present only point estimates for metrics like CLIP and DreamSim. In stochastic deep learning experiments, single-run results are insufficient to claim superiority. The manuscript should report standard deviations over multiple random seeds or training runs and include statistical significance tests (e.g., paired t-tests) to demonstrate that the observed improvements are not due to random variance.

Finally, the user study results in the Appendix report preference percentages without confidence intervals. With 50 test sets and 10 evaluators, the sample size is relatively small for drawing strong statistical conclusions about “significant” superiority. Reporting 95% confidence intervals for these proportions would provide a more honest assessment of the evidence.

Action items.

- **[science]** The VSA metric relies on an MLLM (Qwen-VL-Max) for scoring. The manuscript lacks statistical validation of this evaluator’s reliability (e.g., inter-rater agreement with human judges or consistency across multiple MLLM calls). Without confidence intervals or error bounds for these MLLM-derived scores, the reported improvements (e.g., VSA 4.380 vs 4.150) cannot be statistically distinguished from evaluator noise.
- **[science]** The ablation study (Table 4) presents single-point metric values without measures of variance (standard deviation) or statistical significance testing (e.g., t-tests or ANOVA).

Given the stochastic nature of diffusion training and the small number of ablation runs implied, it is unclear if the observed differences (e.g., CLIP 0.736 vs 0.727) are statistically significant or due to random initialization variance.

- **[science]** The user study (Appendix) reports preference percentages (e.g., 66.2% for Consistency) based on 50 samples and 10 evaluators but omits confidence intervals or binomial test results. The statistical power of this sample size to detect meaningful differences between methods is not addressed, making the claim of ‘significant’ superiority statistically weak.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a generally high level of technical writing, with clear structure and logical flow in most sections. The abstract effectively summarizes the problem and solution, and the introduction clearly outlines the contributions. However, there are several minor grammatical errors and stylistic inconsistencies that, while not severely impeding understanding, detract from the overall polish of the paper.

Specifically, the abstract contains a syntactic error (“numerous these effect LoRAs”) that should be corrected to “these numerous effect LoRAs” or similar. In Section 4.3, a stray space appears in the section title (“Asymmetric Orthogonal Prompting }”), which is a minor formatting oversight. Additionally, in the Introduction’s contribution list, the article “an” is incorrectly used before “zero-shot” (“an zero-shot effect composition”), which should be “a zero-shot”.

The flow in Section 4.4 could be slightly improved. The transition between the definition of Target Simulation and the description of its implementation (“In this branch...”) is somewhat abrupt. A smoother connective phrase would enhance readability. Furthermore, while the claim of reducing deployment overhead to “0.5%” is made in the contributions, the main text’s deployment analysis (Table 2) presents raw storage figures without explicitly calculating or stating the 0.5% figure in the narrative, which might leave the reader to infer the calculation. Explicitly stating the calculation or the basis for the percentage in the text would improve clarity.

Overall, the writing is strong, but addressing these specific grammatical and stylistic points will elevate the manuscript to a higher standard of clarity and professionalism.

Action items.

- **[writing]** In the Abstract, correct the phrase ‘numerous these effect LoRAs’ to ‘these numerous effect LoRAs’ for proper syntax.
- **[writing]** In Section 4.3, remove the trailing space in the title ‘Asymmetric Orthogonal Prompting }’ to fix formatting.
- **[writing]** In Section 4.4, improve the transition between the definition of Target Simulation

and its implementation description to enhance flow.

- **[writing]** In the Introduction, change ‘an zero-shot effect composition’ to ‘a zero-shot effect composition’ to correct the article usage.
- **[writing]** In Section 1, explicitly state the basis for the ‘0.5%’ deployment overhead claim in the text to clarify the calculation for readers.